Question Paper: Machine Learning

Machine Learning Question Paper

Class 9-12

Medium Difficulty

Multiple Choice Questions (1-10)

Choose the correct answer from the given options.

1. What is the primary goal of Machine Learning?

A) To develop software that can understand human language

B) To enable machines to learn from data and make predictions

C) To create robots that can perform tasks autonomously

D) To design algorithms for solving mathematical problems

2. Which of the following is a type of Machine Learning?

A) Rule-based systems

B) Expert systems

C) Supervised Learning

D) All of the above

3. What is the role of the activation function in a neural network?

A) To normalize the input data

B) To introduce non-linearity into the model

C) To reduce the dimensionality of the data

D) To increase the computational complexity

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4. Which algorithm is commonly used for clustering data?

A) Linear Regression

B) Decision Trees

C) K-Means

D) Support Vector Machines

5. What is the difference between overfitting and underfitting?

A) Overfitting occurs when the model is too complex, while underfitting occurs when the model is too simple

B) Overfitting occurs when the model is too simple, while underfitting occurs when the model is too complex

C) Overfitting occurs when the training data is too large, while underfitting occurs when the training data is too small

D) Overfitting occurs when the testing data is too large, while underfitting occurs when the testing data is too small

6. What is the purpose of regularization in Machine Learning?

A) To reduce the impact of outliers on the model

B) To increase the computational complexity of the model

C) To prevent overfitting by adding a penalty term to the loss function

D) To improve the interpretability of the model

7. Which of the following is a characteristic of a good Machine Learning model?

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A) High bias and low variance

B) Low bias and high variance

C) Low bias and low variance

D) High bias and high variance

8. What is the role of the optimizer in a neural network?

A) To update the model parameters during training

B) To regularize the model parameters

C) To normalize the input data

D) To introduce non-linearity into the model

9. Which of the following is a type of neural network?

A) Feedforward Network

B) Recurrent Neural Network

C) Convolutional Neural Network

D) All of the above

10. What is the purpose of cross-validation in Machine Learning?

A) To evaluate the performance of the model on unseen data

B) To increase the computational complexity of the model

C) To reduce the impact of outliers on the model

D) To improve the interpretability of the model

Short Questions (11-15)

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Answer the following questions in 3-5 sentences each.

11. What is the difference between supervised and unsupervised learning? Provide an example of each.

12. Describe the concept of bias-variance tradeoff in Machine Learning. How can it be addressed?

13. What is the role of feature engineering in Machine Learning? Provide an example of a feature engineering technique.

14. Explain the concept of deep learning. How does it differ from traditional Machine Learning?

15. What is the purpose of data preprocessing in Machine Learning? Describe a common data preprocessing technique.

Long/Descriptive Questions (16-18)

Answer the following questions in 8-10 sentences each.

16. Describe the architecture of a convolutional neural network (CNN). How does it differ from a traditional neural network? Provide an example of a real-world application of CNNs.

17. Explain the concept of reinforcement learning. How does it differ from supervised and unsupervised learning? Provide an example of a real-world application of reinforcement learning.

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18. Describe the process of training a machine learning model. What are the key steps involved? How can the performance of the model be evaluated?

Answer Key

1. B) To enable machines to learn from data and make predictions
2. C) Supervised Learning
3. B) To introduce non-linearity into the model
4. C) K-Means
5. A) Overfitting occurs when the model is too complex, while underfitting occurs when the model is too simple
6. C) To prevent overfitting by adding a penalty term to the loss function
7. C) Low bias and low variance
8. A) To update the model parameters during training
9. D) All of the above
10. A) To evaluate the performance of the model on unseen data

11. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data. Example: Image classification (supervised) and clustering (unsupervised).

12. The bias-variance tradeoff refers to the tradeoff between the model's ability to fit the training data (bias) and its ability to generalize to new data (variance). This can be addressed by using techniques such as regularization and cross-validation.

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13. Feature engineering involves selecting and transforming the input data to improve the performance of the model. Example: Normalizing the input data.

14. Deep learning refers to the use of multiple layers in a neural network to learn complex patterns in the data. This differs from traditional Machine Learning, which typically uses a single layer.

15. Data preprocessing involves transforming the input data to improve the performance of the model. Example: Handling missing values.

16. A CNN consists of multiple layers, including convolutional, pooling, and fully connected layers. This differs from a traditional neural network, which typically consists of a single layer. Example: Image classification.

17. Reinforcement learning involves training a model to make decisions based on rewards or penalties. This differs from supervised and unsupervised learning, which involve training a model on labeled or unlabeled data. Example: Robotics.

18. The process of training a machine learning model involves data preprocessing, model selection, training, and evaluation. The key steps involved include data splitting, model training, and model evaluation. The performance of the model can be evaluated using metrics such as accuracy and precision.